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1st SEMESTER 2024/25 Group Coursework

BSc Information Management &
Information Systems

Advanced Database Management

INSTRUCTIONS TO CANDIDATES

1. A single copy of your work must be submitted electronically through Learning Mall
Online by **11:59 pm, Friday, 18 Oct. 2024** (China Standard Time, UTC/ GMT+8).
2. This coursework will count for 30% of the final mark in this course.

Coursework Title:

Group coursework: Designing a Database

Scenario

Context

UniBooks is a small start-up company specialising in second-hand textbooks. The company buys surplus academic books in bulk from publishers but also buys individual books from students. The books are advertised and sold via the company's website. **You have been approached to develop a database system for the company.** The system needs to be able to store details of the books held in stock, to process incoming books and to record all sales. The system will also need to be able to perform other functions, such as producing invoices and different management reports.

Specification

UniBooks supplies textbooks to university students at discounted prices. The company obtains stocks of books directly from publishers and students:

- Usually, UniBooks is able to purchase large stocks of books from publishers for as little as 30% of the cover price.
- After completing a course, a student may wish to sell his/her unwanted textbooks. UniBooks buys textbooks directly from students for 20% of the cover price.
- All books are sold to the public for 80% of the cover price. There is also a standard charge of ¥10 for postage and packaging, regardless of the size of the order placed.

The current system suffers from several limitations:

- Occasionally, a customer will order a book that is not held in stock (e.g. print on demand titles). A staff member will need to search for details of the book's publisher, and then contact the publisher to see if UniBooks can obtain a copy of the book. The current system does not provide a search facility that can be used to locate a publisher's details.
- When stocks of a given book run out, it is left to a staff member to reorder a fresh supply from publishers. This raises two problems: customer orders are delayed until new stocks of the book arrive, and there is inconsistency regarding the number of books ordered from publishers. As an example, sometimes too few or too many books are ordered, resulting in further delays to customers or increased inventory costs. Ideally, it should be possible to set reorder levels and reorder quantities for books to improve inventory control.
- Management information takes a great deal of time to produce since it is collected/processed manually. As a result, managers are only able to see important information, such as sales figures, on a monthly basis. Ideally, management information should be available on demand.
- Most publishers offer discounts based on order size (in addition to the discount over cover price). In general, discounts are given as shown below in Table 1. At present, UniBooks often fails to claim the discounts it is entitled to.

Table 1: Discount from Publishers

Copies Ordered	Discount (% Of Total Order Value)
20 - 49	5

50 - 99	7.5
100 - 199	10
More than 200	12.5

- UniBooks offer discounts to customers based on order size (in addition to the discount over cover price). Currently, these discounts are processed manually, and the sales information is not recorded accurately. In general, discounts are given as shown below in Table 2.

Table 2: Discount to buyers

Order Size	Discount (% Of Total Order Value)
10 -19	5
20 - 29	7.5
30 - 39	10
More than 40	12.5

Task Details/Description:

The primary task for your group is to go through the scenario context and specification to understand and analyse the company's data and information requirements. This should help you to design an appropriate database structure, which can meet the requirements of the organisation. In developing, realising and implementing the design, your group must create sample records (dummy data) for the identified entities.

This is a group task and each member of the group must complete an equal share of the work. Each group will be asked to review its own performance and may be given an opportunity to redistribute some of the marks awarded for the work (in case of conflicts within the group members). A list of tasks (not exhaustive) that you will need to perform as a group in completing this coursework are given below.

Tasks:

- **TG1:** Decide upon *the entities* to create tables and corresponding attributes you will need, to address a range of requirements stemming from the scenario and specification. Example entities and this is not an exhaustive list: customer, book, student, employee, order etc.
- **TG2:** Create an *Entity-Relationship Model (ERM)* using Microsoft Access to address the requirements. This model must be *normalised* to reduce data redundancy and improve data integrity.
- **TG3:** Include *sample records* (at least **four**) in Microsoft Access for each table after developing the ERM and include validation of the attribute types, constraints and primary/foreign keys.
- **TG4:** Reflect upon your ERM, discuss **two** *issues/limitations* associated with the design and anticipated operation of your database, and its impact on the business, i.e. how it might affect the business.

All these above the tasks will help you to make progress and completing this group coursework. Upon completion of the group coursework, the deliverables produced will provide a suitable starting point for the individual coursework. Additionally, feedback provided on the group coursework will aid in refining the work you will undertake in the individual coursework

Please ensure you have created one or more back-ups of your work in suitable places using appropriate mediums available to you, to avoid losing the work you may have already done.

Your work will be assessed on robustness, correctness, design meeting the business requirements, and the overall quality of your design.

Deliverables:

You **MUST** submit the following to complete this coursework.

- **Deliverable 1:** The **report (as a word or PDF document)** should not exceed 500 words (+/- 10%). Estimated word limits for each section are provided below to help you write the report.
- **Deliverable 2:** **Entity Relationship Model** as a PDF document from Microsoft Access (*Further instructions on how to save Entity Relationship Model as a PDF from Microsoft Access will be uploaded in blackboard in due time*)
- **Deliverable 3:** Your **Microsoft Access Database File**

Contents of the report [deliverable 1]

- **GD1:** The **front cover page** of the report must include your group number and student number for each member in the group. [*No word count applies*]
- **GD2:** Briefly explain the normalisation and resolving m:m relationship process with the aid of a suitable example (for each part – may use the same example) drawn from your own ER model. You should not explain normalisation and its objectives, rather explain how you have employed normalisation and resolved m:m relationship in your ER model (using suitable at least one suitable example). [*estimated word limit – not more than 300 words*]
- **GD3:** A brief discussion of **two issues** associated with the design and anticipated operation of your database, and how these issues may impact the business/organisation. You may discuss any issues/problems for which a solution is not yet clear, or where there are competing alternative designs. [*Estimated word limit – not more than 200 words*]. **Please note, you MUST at a (or the) minimum discuss two issues suitably in your report (as stated above) to score a passing mark for this component.**

Note

1. Please refer to the grading criteria at the end of this document, to know how each component or a combination of components will be graded.
2. You will submit three separate files electronically. Further instructions about the submission will be provided in Week 4 lecture.

Feedback Methods:

Formative feedback will be available before the final submission, to help you develop and refine your work. This will include the following methods:

- **Weekly labs** – these will include tasks and activities (related to the coursework) to help you apply knowledge from the lectures, develop your own solution and obtain feedback in the session to help you to improve your work. You are therefore encouraged to engage during the lectures and workshops to make progress, receive feedback, clarify doubts, and refine your work.
- **Assignment clinic** – In week 5, there will be sessions dedicated to reviewing the assessment requirements, clarify areas of doubt and obtain further feedback on your work, before the final submission.
- **Office hours** – You are advised to use office hours early in the term, so you have time to act on the feedback.
- **Discussion Forum in LM** – you can ask questions, discuss among peers and receive responses via the discussion forum. Turnaround time for a response from the module tutor is 2-3 working days or it will be given in the following lecture.

Summative feedback will be provided after the final submission. This accompanies your final grade and uses the assessment criteria shown at the end of this document.

Submission:

The coursework is going to be submitted electronically. Further instructions about how to do this will be given later.

You will submit 3 files including:

- **Group report** (either as a pdf or word document)
- **Entity relationship model** (as a pdf saved from Microsoft Access)
- **Microsoft Access database file.**

Assessment Weighting for the Module:

The weight of the group coursework is **30%**

Assessment Criteria

Marking guidelines for this coursework are provided at the end of this document.

Learning Sessions Relevant to this Coursework

- Week 1: Introduction to Databases
- Week 2: Entity Relationship Modelling
- Week 3: Normalisation
- Week 4: Recap
- Week 5: Formative Feedback Session

Contact

If you have any questions regarding this coursework, please email to ou.liu@xjtlu.edu.cn or Wenzhe.Liang19@student.xjtlu.edu.cn, OR you may book office hours with the teaching team.

Group Issues

You have the chance to form your group of 4 or 5 persons in week 1 & 2. At the end of week 2 class, if you have difficulty to form a group or your group size is less than normal size, we will help you randomly allocate more groupmates to make up the groups. The group list will be post in week 3. It is your responsibility to get in touch with group members and do the work with your group.

Group Coursework Marking Guidelines

	70% and above	60%-69%	50%-59%	40%-49%	Less than 40%
ER Model (50%) Accompanying submission [PDF file of the ER model]	ER Model meets and exceeds the requirements in the specification. The model is fully normalised and uses a comprehensive and realistic set of entities. All relationships and cardinalities have been correctly identified. The model uses suitable naming conventions, notations and is presented in a clear and professional way. The work demonstrates excellent design that meets and exceeds the business needs, realised using normalised entity-relationship modelling.	ER Model meets most requirements in the specification. The model is mostly normalised and uses a realistic set of entities. Most relationships and cardinalities have been correctly identified. The model mostly uses suitable naming conventions, notations and is presented clearly. The work demonstrates very good design that meets most of the business needs, realised using normalised entity-relationship modelling.	ER Model meets some requirements in the specification. The model is partially normalised and uses a limited set of entities. Some relationships and cardinalities have been correctly identified. The model generally uses suitable naming conventions, notations but may lack clarity in presentation. The work demonstrates a good design that meets some of the business needs, realised using normalised entity-relationship modelling.	ER Model meets few requirements in the specification. The model is not normalised and uses very limited set of entities. Most relationships and cardinalities have not been correctly identified. The model lacks suitable naming conventions, notations and presentation. The work demonstrates a design with limitations that meets few of the business needs, realised using normalised entity-relationship modelling.	ER Model is either incomplete or missing. The model is not normalised, and entities have not been correctly identified. The work demonstrates incomplete design with several limitations that does not meet the business needs.
Tables and Sample Records (25%) Accompanying submission [Microsoft Access database file]	All entities have been identified with a realistic set of attributes, meeting and exceeding the business needs. The attribute types, validation rules and constraints for all the entities have been identified correctly. Sample records for each entity demonstrates excellent understanding of the entities and relationships between them. Work demonstrates excellent understanding of the essential concepts, and process required to design databases conforming to user requirements.	Most entities have been identified with a realistic set of attributes, meeting the business needs. The attribute types, validation rules and constraints for most entities have been identified correctly. Sample records for each entity demonstrates very good understanding of the entities and relationships between them. Work demonstrates very good understanding of the essential concepts, and process required to design databases conforming to user requirements.	Several entities have been identified with a set of attributes, meeting some of the business needs. The attribute types, validation rules and constraints for some entities have been identified correctly. Sample records for each entity demonstrates limited understanding of the entities involved in ERM. Work demonstrates good understanding of the essential concepts, and process required to design databases conforming to user requirements.	Few entities have been identified with a set of attributes, meeting few of the business needs. The attribute types, validation rules and constraints for some of these entities have been identified correctly, and mostly incomplete. Sample records are mostly incomplete and demonstrates very limited understanding of the entities and relationships between them. Work demonstrates limited understanding of the essential concepts, and process required to design databases	Very few entities have been identified with a set of attributes, meeting the business needs. The attribute types, validation rules and constraints for these entities have not been identified correctly, and mostly incomplete. Sample records are missing. Work demonstrates very limited understanding of the

				conforming to user requirements	essential concepts, and process required to design databases conforming to user requirements
Report (25%) Accompanying submission [report as a word or PDF document]	Very well structured and presented. Effectively and convincingly communicates to a professional standard. Writing is very clear and concise, flows logically, leaves no room for reduction and contains no (or only minor) spelling/grammatical errors. Report demonstrates excellent understanding of key design techniques, the limitations of the design, its impact on the business and key information needs of the organisation.	Well-structured and presented that communicates effectively. Writing is clear and concise, flows logically, and is largely free of spelling/grammatical errors. Report demonstrates very good understanding of key design techniques, the limitations of the design, its impact on the business and key information needs of the organisation.	Communication is generally clear but may lack a logical structure. Writing contains spelling/grammatical errors that affect clarity. Report demonstrates good understanding of key design techniques, the limitations of the design, its impact on the business and key information needs of the organisation.	A lack of clear structure and spelling/grammatical errors make the work difficult to read, but some logic is evident. Report demonstrates limited understanding of key design techniques, the limitations of the design, its impact on the business and key information needs of the organisation.	Very disorganised and unclear. Writing is unprofessional or contradictory and does not coherently explain the writer's logic. Report demonstrates almost no understanding of key design techniques, the limitations of the design, its impact on the business and key information needs of the organisation.